Master Laboratory: Cryopreservation of Cell Suspensions

1 Introduction

1.1 Cryopreservation

Cryopreservation is the use of low temperatures to preserve the structural and functional integrity of cells and tissues for long periods of time [1]. Cryopreservation refers to the process of freezing biological material at extremely low temperatures (from −80°C). Liquid nitrogen is often used to achieve temperatures as low as −196°C. The aim is to stop biological activity, including the biochemical reactions that lead to cell death and DNA damage, at low temperatures. This allows biological material to be preserved for weeks or even years, with the possibility of restoring cell functionality after thawing [2].

The use of cold to preserve biological material began in earlier civilisations, and traces can be found in Mesopotamia and in almost all civilisations that had access to ice. Experiments with the preservative effect of cold began in the 17th century. The experimenter Boyle tried to preserve whole animal bodies in ice by freezing them and trying to revive them. Boyle also discovered species of frogs and fish that could survive in the wild [3].

In the 18th century, after the development of the microscope, Spallanzani discovered that sperm could retain their motility at low temperatures. However, further studies on the cryopreservation of sperm and erythrocytes showed that this method was unstable [4]. In the last century, the field of cryopreservation research has grown in importance. There were two main developments that led to the new generation of cryogenic research.

Chronologically, the first was the discovery of the cryoprotective properties of glycerol. The year 1949 is considered by many researchers to be the turning point in the history of cryobiology. In that year, Polge, Smith and Parkes published a scientific paper describing the cryoprotective properties of glycerol. This is considered a milestone in the field of cryobiology [5]. A second milestone was the observations of James Lovelock in the 1950s. He described the formation of ice crystals during freezing and the resulting osmotic pressure.

Later, in 1963, Mazur was able to describe this process in more detail and prove that the cooling rate of a cell suspension controls the movement of water across the cell membrane and thus the degree of intracellular freezing. He developed the "2-factor hypothesis". In the 1980s, research into the cryopreservation process showed that the rate of freezing and thawing was the most important factor in cell survival. It was shown that small, slow steps in both the freezing and thawing process prevented the rapid formation of ice crystals [4].

Cryopreservation has since become established as a practical method of storing living cells and tissues, widely used in medicine and biology [3]. It is an important technology that is essential for many cell therapies, such as stem cell therapy, tissue engineering or assisted reproduction, because cell therapies often require the preservation of cells or specific cellular components at several points in the manufacturing process. A Food and Drug Administration (FDA) publication found that more than 80% of stem cell therapy applications use cryopreservation as part of the manufacturing process for storage and delivery of their products [6].

1.2 Mechanisms of cell death in cryopreservation processes

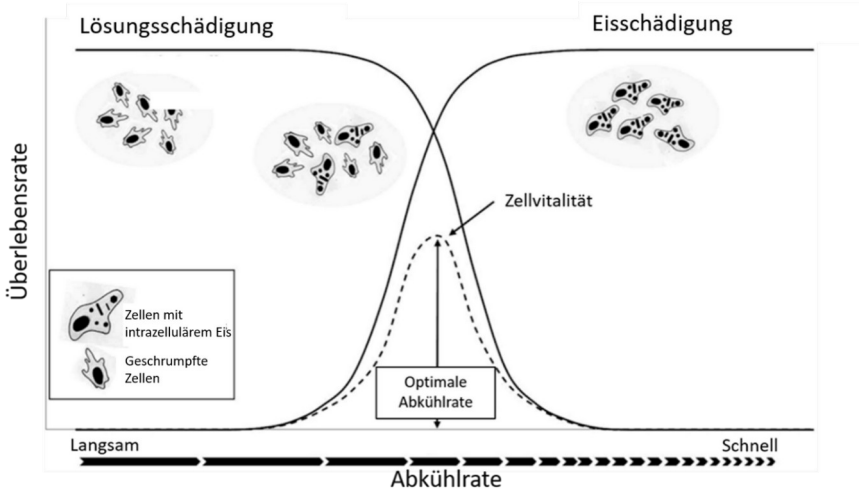
Successful cryopreservation depends on several factors. Cooling cells or organs often causes irreversible damage to cell membranes or cells. However, the advances in research mentioned above have led to a significant improvement in the process. The understanding of the destructive

mechanisms has made it possible to avoid them to a large extent by optimising the method. This has led to optimised preservation protocols that require different tools. In order to understand the roles and effects of these tools, the mechanisms of cell death during cooling are first explained here.

First of all, it is important to know that cooling biological material below 0℃ is usually fatal. The main problem is water, which makes up about 80% of biological material. Since the freezing of water means the transition to the solid aggregate state, which then occurs in the form of ice crystals, the first fatal effect can be explained purely mechanically. Intracellular ice crystals can pierce or rupture the cell membrane and cause cell death. Extracellular ice crystal formation is also problematic from a biochemical point of view. Extracellular ice crystals increase the electrolyte concentration in the extracellular space. As a result, osmotic pressure increases and intracellular water penetrates the cell membrane to balance the electrolyte concentration. Gradually, this mechanism leads to cell shrinkage, which can even lead to cell death [1]. A distinction has therefore been made between two types of cell damage: Ice damage is a mechanical injury caused by the formation of ice crystals that rupture the cell membrane. The second type, known as solution damage, is caused by osmotic pressure, which leads to cell shrinkage [7].

Two main measures are taken to eliminate these mechanisms. Firstly, the cooling and heating rates and the temperature at which ice crystals form determine the outcome of cryopreservation. The cooling rate has a direct effect on cell dehydration and therefore on solution damage during freezing. In addition, the rate of water transport across the cell membrane depends on the temperature, which has an indirect effect on the probability of lethal intracellular ice formation or ice damage. Thus, based on Mazur's 2-factor hypothesis of cryodamage (Figure 1), there is an optimal cooling rate that should be achieved during the cryopreservation process. The controlled freezing rate therefore determines the degree of dehydration during freezing and thus the survival rate of the cells.

The hypothesis also assumes that the optimal cooling rate is different for different cell types and corresponds to the rate at which the osmotically active volume of the cells at –30℃ decreases to 90-95% of the original volume. In addition, the addition of cryoprotectants plays a critical role in the survival rate of frozen cells and results in the optimal cooling rate being spread over a wider range. This range depends on the type and concentration of cryoprotectant used. Ultimately, the survival rate is determined by a complex interplay between the cooling rate and the type and concentration of



cryoprotectant [8].

Figure 1: The 2-factor hypothesis: Left: At low cooling rates, the probability of solution damage increases. Right: Fast cooling rates increase the probability of ice damage [1]

1.3 Cryoprotective agents

As mentioned above, cryoprotectants (CPAs) are used to prevent cell damage during the cryopreservation process. Inspired by the so-called antifreeze proteins (AFPs) found in polar fish and other animals to overcome low temperatures, cryoprotectants are also of paramount importance in cryobiology. In general, cryoprotectants can be defined as a group of substances that can be used to minimise or, at best, prevent the damage caused by freezing or thawing [9] [10].

Cryoprotectants play a central role in protecting cells from cryo-damage and allowing them to be successfully stored at cryogenic temperatures. They fall into two main categories: Penetrating and non-penetrating cryoprotectants, depending on whether or not they penetrate the cell membrane [7]. Penetrating cryoprotectants are not only distributed in the extracellular space, but also penetrate the cell membrane and are therefore inside the cells. This property is mainly due to the size of these molecules and has the advantage of reducing intracellular ice crystal formation. Examples include DMSO and glycerol. Non-penetrating cryoprotectants have larger molecules and are unable to penetrate the cell membrane. They therefore only move in the extracellular space. Trehalose or sucrose, for example, are used as non-penetrating cryoprotectants [4].

To better understand the protective mechanisms of cryoprotectants, it is important to know the structure of intracellular water. Different water compartments are thought to exist within cells, containing both freely diffusible (and freezable) water and restrictedly mobile or bound water [11]. The addition of a penetrating cryoprotectant, such as DMSO, increases the amount of bound water. Such cryoprotectants form hydrogen bonds when in contact with water. This minimises the formation of intracellular ice crystals and the resulting dehydration due to heat dissipation. However, they are generally toxic and, at high concentrations, can cause harmful changes in the genetic material of cells. Non-penetrating cryoprotectants, such as trehalose or sucrose, remain in the extracellular space and can regulate, deform and delay the formation of ice crystals. They can also inhibit dehydration and recrystallisation of ice during thawing, but are less effective than penetrating agents. In order to take advantage of both types of CPA and to reduce the concentrations and thus the toxicity of penetrating cryoprotectants, mixtures of both types are often used [12].

1.4 Technique for freezing, storing and thawing

As mentioned above, the aim of cryopreservation is to cool the material to be preserved to very low temperatures at a defined rate. This can be achieved using passive cooling devices or freezers with a controllable cooling rate. Passive cooling devices (PCDs) or freezers are an economical option. The principle is to control the heat dissipation after positioning in a mechanical freezer (−80℃) and to ensure a cooling rate of −1 K/min. Control is achieved either by adding a heat transfer medium such as isopropanol (e.g. Mr Frosty, Nalgene) or an insulating foam (e.g. CoolCell, BioCision). The rate of approximately −1 K/min must be guaranteed, especially in the temperature range −10℃ to −40℃ [1].

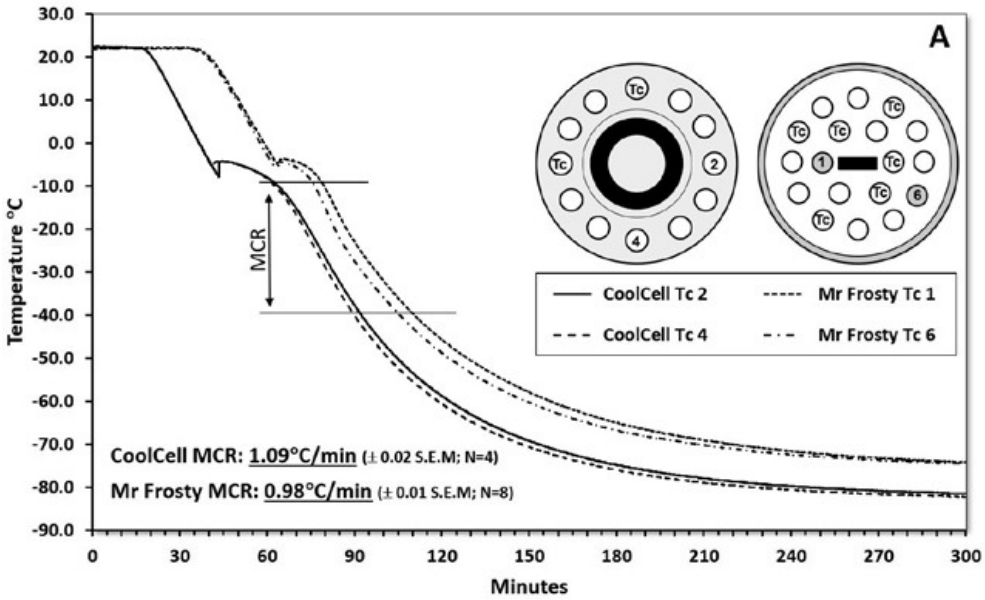


Figure 2: Cooling rate of PCD [6]

The Controlled Rate Freezers (CRF) use liquid nitrogen (LN2) as the refrigerant and control the cooling rate via a solenoid valve which, in conjunction with a heating coil, regulates the temperature in the sample chamber. These units are programmable and can cool down to −100℃ and provide variable multi-phase linear cooling rates. In addition, some CRFs are operated without LN2, using a Stirling cycle and heat pumps. However, these can only achieve a maximum cooling rate of −2 K/min [1]. As they offer better control over the cooling rate and allow it to be set flexibly, CRFs are recommended in the context of cell therapy [6].

With regard to storage, the ideal storage temperature of a cell suspension is the glass transition temperature (Tg), which is, for example, −132.58℃ for a cell suspension containing 10% DMSO by weight. If the temperature of a solution falls below the Tg, the mobility of the water molecules is reduced due to the increased viscosity. Therefore, cell suspensions should be stored at a temperature below which they are completely vitrified and no metabolic processes can take place. It is also possible to store some cell types in freezers with a storage temperature of −80℃. Red blood cells can be stored for 10 years under these conditions when cryopreserved in 40% glycerol [13] pp 65-69. However, this is not recommended for cell therapy purposes and storage at ≤ −130℃ is recommended [14].

The thawing of cells is not as important as freezing and has therefore not been studied to the same extent. It is generally assumed that cells must be heated rapidly to prevent recrystallisation of ice. Typically, samples are thawed by placing them in a water bath at 37℃ for one and a half to two minutes [4].

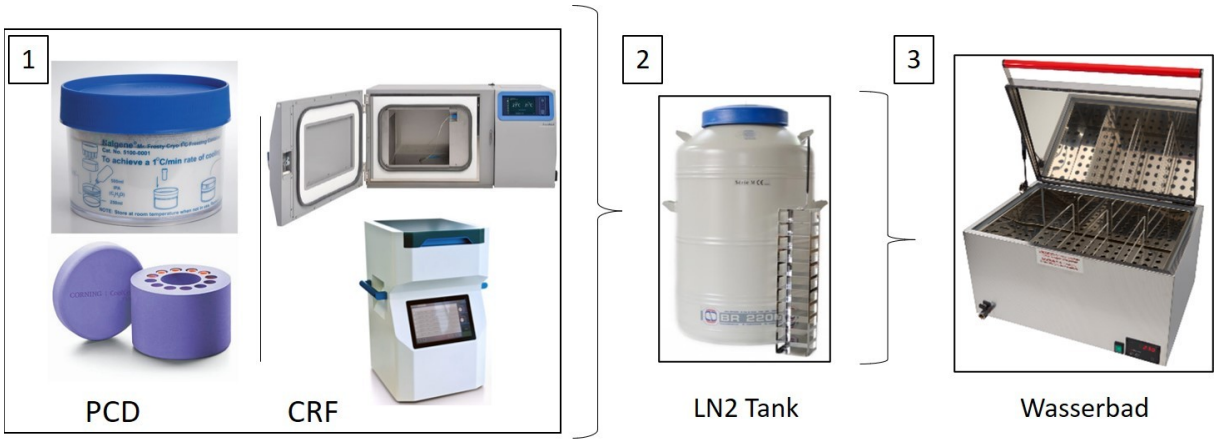


Figure 3: Devices for freezing, storing and thawing; 1) the freezers, on the left are the passive freezing containers, top: Mr.Frosty, Nalgene® [15] bottom: coolcell, Corning® [16]. On the right are the cooling units with a controllable cooling rate, above: LN2 freezer CryoMed, Thermo Scientific [17], below: A Stirling cycle freezer VIA FREEZ, Cytiva [18]. 2) Liquid nitrogen tank for storing cryopreserved samples with a BR2000 stand, CRYOGENIC [19]. 3) Water bath for thawing the samples [20].

1.5 HeLa cells

HeLa cells are an immortal human cell line derived from a cervical carcinoma and are the first human cells successfully maintained to divide indefinitely in vitro, making them among the most widely used model systems in biomedical research. HeLa cells proliferate unusually fast and express an active form of telomerase, which maintains telomere length and allows them to bypass the limit and cellular senescence.

In routine cell culture, HeLa cells and stem cells (MSCs) share basic handling principles, but stem cells need stricter control to maintain their undifferentiated state and specialised functions. For a more in-depth exploration of cell culture, watch the video at https://av.tib.eu/media/58 271 This video provides detailed insights into the methods of cryopreservation of cells. It serves as a valuable resource to complement your theoretical understanding and practical skills for this Master Lab.

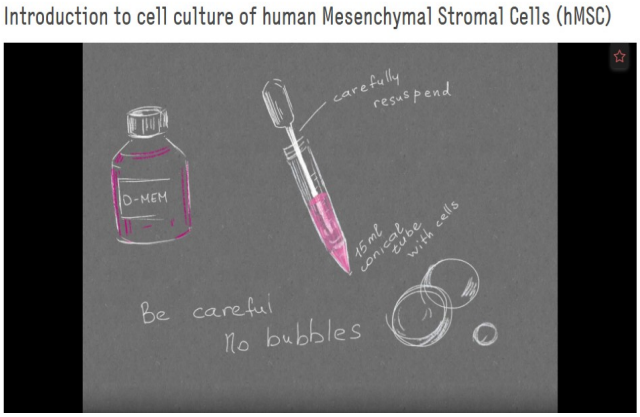


Figure 4: Cover video von IMP-TIB for cell culture of human Mesenchymal Stromal cells.

2 Materials and methods

For all students working in this laboratory: Wear lab coats, wash and disinfect hands, tie up hair, wear gloves, clean gloves and work surfaces with 70% ethanol. All required materials will be found under the cell culture workbench and are properly labelled.

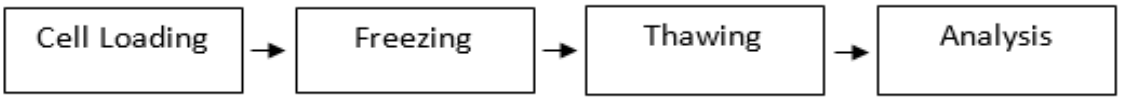
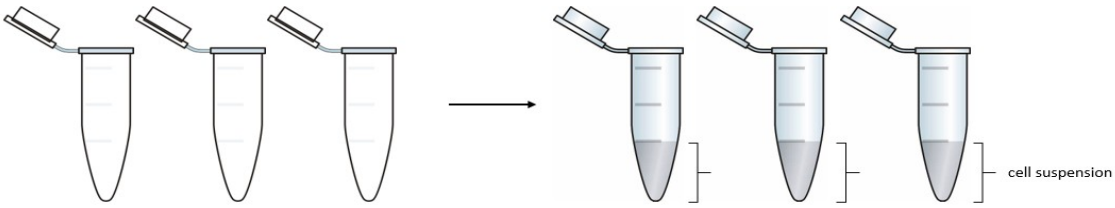
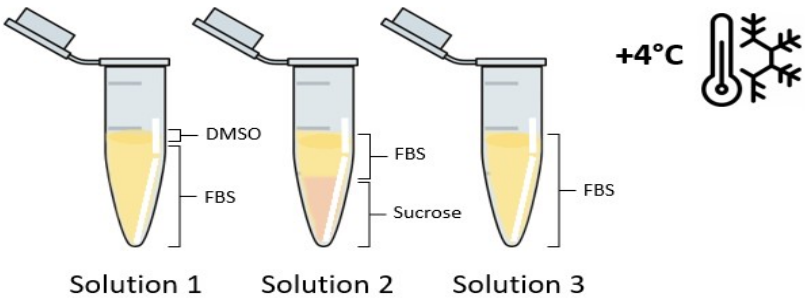


Figure 5: Working process for the cryopreservation of HeLa cells

1. Take 3 Eppendorf tubes (2 ml) and fill them with 500µl of the cell suspension.

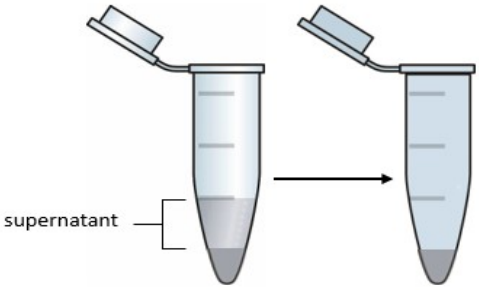


2. Centrifuge at 1600 RCF for 5min.
3. Turn ON the Asymptote to start the freezing protokol. This will be cooling to 4℃, hold there until loading the samples and afterwards cooling at 1K/min until −40℃
4. Prepare the CPA solutions in 3 different Eppendorf tubes. Be sure to keep all of them on ice during preparation.

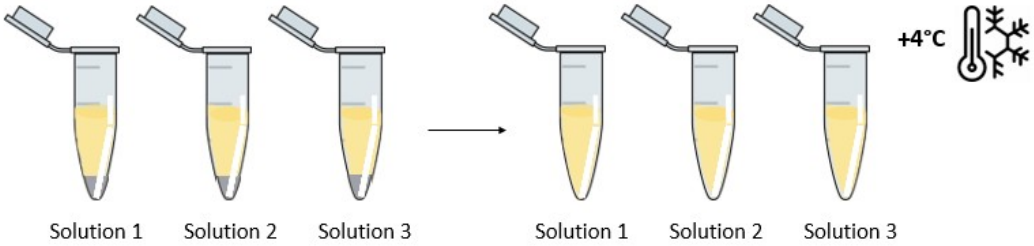


	DMSO	1M sucrose	FBS	PBS
Solution 1	50µl	-	950µl	-
Solution 2	-	500µl	500µl	-
Solution 3	-	-	-	1000µl

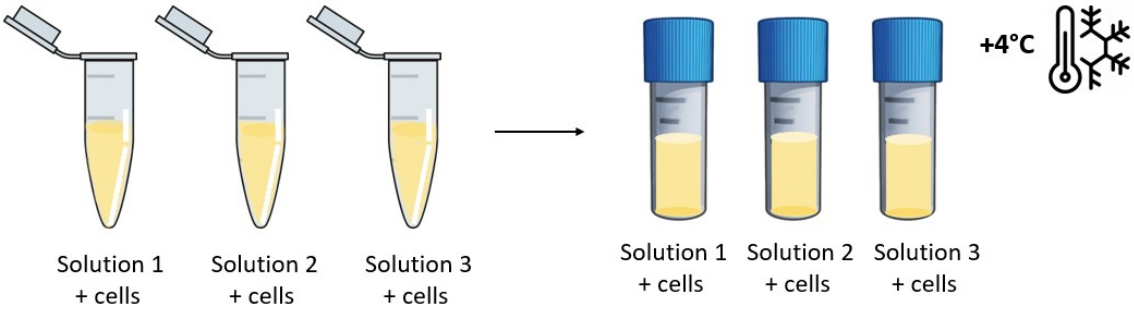
5. Take the 3 Eppedorf from the centrigufe and aspirate the supernatant. Be careful not to aspirate cells.



6. Pipette each of the CPA solutions in one of the eppedorfs with cells and slowly resuspend. Keep the cells and the CPAs on ice.

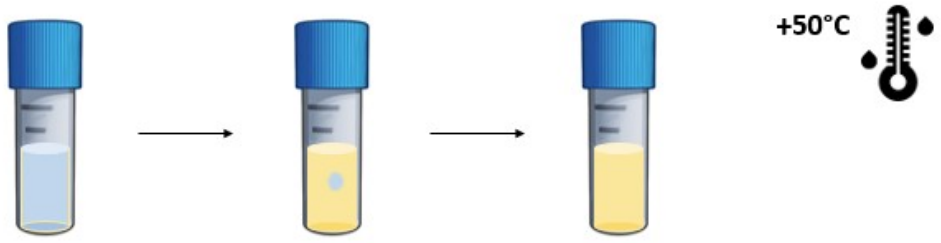


7. Transfers each of the 3 solutions with cells to cryovials and keep them in ice for around 10 min.

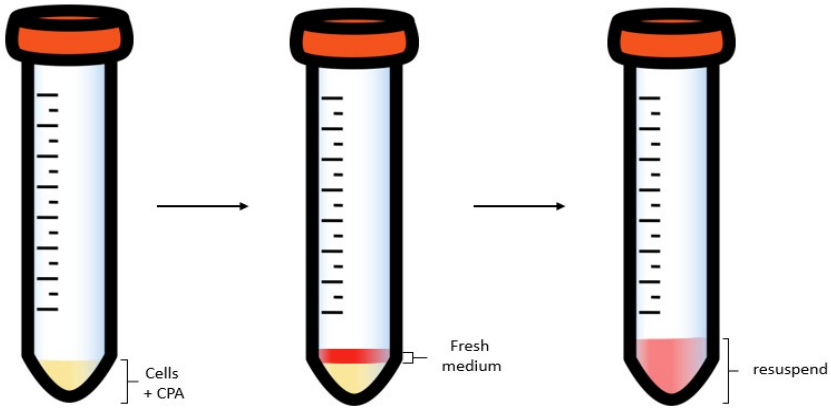


8. Transfer the cryovials to the Asymptote and start the freezing protocol.
9. Turn on the water bath and set it at 50℃ to prepare for the thawing step.
10. Take the cryovials from the asymptote one by one and thaw them.

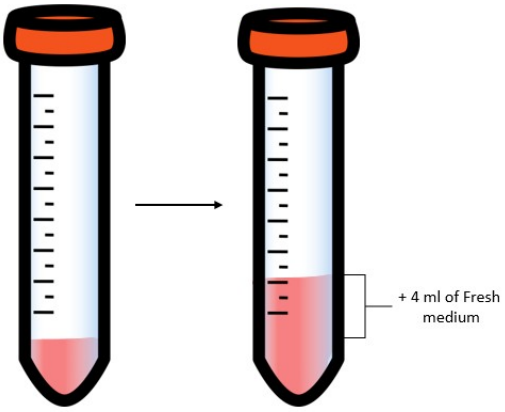
11. Use tweezers to immerse the cryovial in the water bath at 50℃. Move the cryovial until a little piece of ice is left.



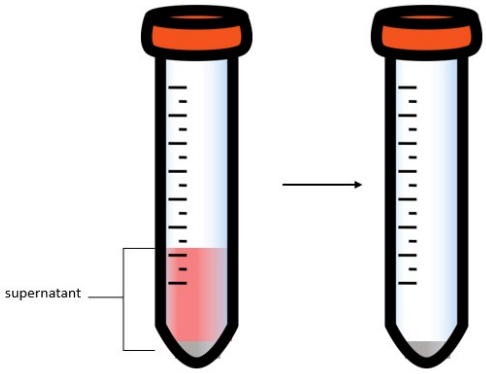
12. Transfer the cryovial to the working bench.
13. Take 15ml falcon tube and transfer the cryovial content to the tube. Wash the cryovial with 1 ml medium. Use medium at room temperature for this and next step



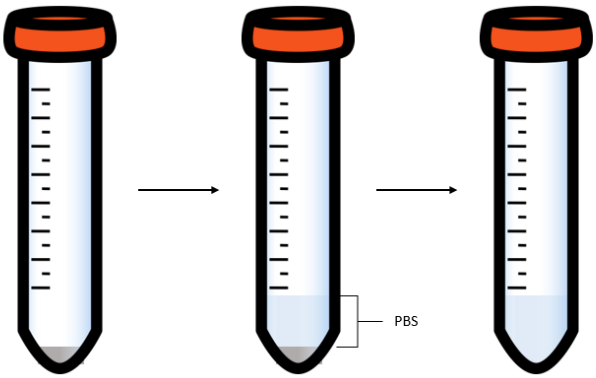
14. Add more medium in the falcon tube to complete 5ml medium. It is required 4ml medium per 1ml of DMSO. In this step the CPA is unload.



15. Centrifuge the falcon tubes at 1000RPM for 5 min.
16. Take the falcon tubes from the centrigufe and aspirate the supernatant. Be careful not to aspirate cells.



17. Add 3ml of PBS to each falcon tube and resuspend the cells.



18. Transfer 1ml of the solution to the vicell cuvettes. Prepare 3 cuvettes per tested solution.
19. Transfer the cuvettes to vicell and count them.
20. Save the obtained results and use them for the evaluation.

3 Results and Analysis

Note: For all the analysis, when sample values have been taken in triplicate report the value as the median ± standard deviation.

3.1 Calculation recovery rate

Using the data obtained from the Vicell calculate and report the recovery rate for each of the samples frozen and thaw with the different solutions.

To calculate the recovery rate after freezing cells, you can use the following formula (1):

% Recovery = [(# of cells recovered) x 100] / [(# of cells frozen)] (1)

3.2 Calculation freezing rate

Using the data obtained from the Asymptote make the graph temperature vs time for each of the solutions. Calculate the freezing rate and compare the value with 1K/min.

To calculate the freezing rate from a curve of time versus temperature, you can use the following formula (2):

Freezing Rate = $\Delta t/\Delta T$ (2)

where ΔT is the change in temperature and Δt is the change in time. You can calculate

ΔT = final temperature – initial temperature

Δt = final time – initial time

Make two calculations one until the final freezing temperature (–80℃) and the second one until the nucleation temperature.

3.3 Calculation Thawing rate

Used the values recorded to calculate the thawing rate, you can use the following formula (3):

$$\text{Thawing Rate} = \Delta t/\Delta T \tag{3}$$

where ΔT is the change in temperature and Δt is the change in time. You can calculate

ΔT = final temperature – initial temperature

Δt = final time – initial time or total thawing time

3.4 Final presentation

Prepare per group a final presentation including, introduction, materials and methods, results and discussion, conclusions. The total presentation time will be 15 min per team and all participants of the team should present a part of the presentation. At the end each group will have 5 to 10 min for questions.

For discussion the following questions should be addressed:

1. What are the most common cryoprotectants used for HeLa cells?
2. What is the optimal cooling rate for cryopreserving HeLa cells?
3. What are the most common methods for thawing cryopreserved HeLa cells?
4. What is the effect of cryopreservation on the viability and functionality of HeLa cells?

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